

(*) All the rings should be regarded as commutative rings with identity if there is no special statement. In this section, M will be assumed to be an A -module.

Nakayama Lemma

Theorem 0.1. :Let I be an ideal of A and $I \subseteq J(A)$, where $J(A)$ is the Jacobson radical of A . Let M be a finitely generated A -module, if $M = IM$, then $M = 0$.

We prove a conclusion that will be used later

For any ideal I of A , we have $I(M/N) = (IM + N)/N$.

Proof: Consider the natural homomorphism $\varphi : M \rightarrow M/N$, since $I(M/N)$ is a submodule of M/N , we have $N \subseteq \varphi^{-1}(I(M/N))$. We can know that $I(M/N) = \{\sum_i a_i \overline{m_i} | a_i \in I, m_i \in M\} = \{\sum_i \overline{a_i m_i} | a_i \in I, m_i \in M\}$, therefore $IM + N \subseteq \varphi^{-1}(I(M/N))$. On the other hand, for $\forall \sum_i a_i \overline{m_i} = \sum_i \overline{a_i m_i} \in I(M/N)$, we have $\varphi^{-1}(\sum_i a_i \overline{m_i}) = \{\sum_i a_i m_i + n | n \in N\} = \sum_i a_i m_i + N \in IM + N$, which means that $\varphi^{-1}(I(M/N)) = IM + N$, hence $\varphi^{-1}(I(M/N)) = IM + N$. $\square$

There is a useful corollary from Nakayama Lemma

Corollary 0.1. :Let A be a local ring with the unique maximal ideal $\mathfrak{m}$. Let M be a finitely generated A -module. If N is a submodule of M such that $M = N + \mathfrak{m}M$, then $N = M$.

Proof: Consider the A -module M/N , here $J(A) = \mathfrak{m}$. By the conclusion above, we get $\mathfrak{m}(M/N) = (\mathfrak{m}M + N)/N = M/N$, by Nakayama Lemma, we have $M/N = 0$, i.e., $M = N$. $\square$

Let M be a finitely generated A -module and I is an ideal of A , then M/IM is a finitely generated A -module. Indeed, if $\{m_1, m_2, \dots, m_n\}$ is a generating set for M , then $\{\overline{m_1}, \overline{m_2}, \dots, \overline{m_n}\}$ is a generating set for M/IM . Further more, M/IM is a finitely generated A/I -module with the scalar multiplication

$$A/I \times M/IM \rightarrow M/IM$$

$$(a + I, m + IM) \mapsto am + IM$$

it can be checked that such defined scalar multiplication is well-defined. $\{\overline{m_1}, \overline{m_2}, \dots, \overline{m_n}\}$ is also a generating set for the A/I -module M/IM .

Proposition 0.1. : Let A be a local ring with the unique maximal ideal $\mathfrak{m}$ and M is a finitely generated A -module. By the discussion above, we know that $M/(\mathfrak{m}M)$ is a finite dimensional vector space over the field $A/\mathfrak{m}$. Let $\{\overline{m_1}, \overline{m_2}, \dots, \overline{m_n}\}$ be an $A/\mathfrak{m}$ -basis for $M/(\mathfrak{m}M)$ and let m_i be preimage of $\overline{m_i}$. Suppose N is the submodule generated by $\{m_1, m_2, \dots, m_n\}$, then $N = M$.

Proof: It's clear that $N \subseteq M$, let $\varphi : M \rightarrow M/(\mathfrak{m}M)$ be the natural homomorphism, then $\varphi(N) = (N + \mathfrak{m}M)/(\mathfrak{m}M)$. Note that $\{\varphi(m_1), \varphi(m_2), \dots, \varphi(m_n)\} = \{\overline{m_1}, \overline{m_2}, \dots, \overline{m_n}\}$ also generates $M/(\mathfrak{m}M)$, hence we get $\varphi(N) = M/(\mathfrak{m}M)$, therefore $M/(\mathfrak{m}M) = (N + \mathfrak{m}M)/(\mathfrak{m}M)$. Then

$$\frac{M/(\mathfrak{m}M)}{(N + \mathfrak{m}M)/(\mathfrak{m}M)} \cong M/(N + \mathfrak{m}M) = 0,$$

which means $M = N + \mathfrak{m}M$. By corollary 0.1, we have $M = N$. $\square$

recall the definition of exact sequence

Definition 0.1. : Suppose

$$\dots \longrightarrow M_i \xrightarrow{f_i} M_{i-1} \xrightarrow{f_{i-1}} M_{i-2} \longrightarrow \dots$$

is a sequence of A -modules and A -module homomorphisms. This sequence is called **exact** if $\text{Im} f_i = \ker f_{i-1}$ ($\forall i \in \mathbb{Z}$).

Example 0.1. :

$$0 \longrightarrow N \xrightarrow{f} M$$

is exact $\Leftrightarrow \ker f = \{0\} \Leftrightarrow f$ is injective.

Example 0.2. :

$$N \xrightarrow{g} M \longrightarrow 0$$

is exact $\Leftrightarrow \text{Im} g = M \Leftrightarrow g$ is surjective.

Hom functor is left exact

Proposition 0.2. : Let N be an A -module, let

$$M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$$

be an exact sequence of A -modules. Then we have the A -module exact sequence

$$0 \longrightarrow \text{Hom}_A(M'', N) \xrightarrow{g^*} \text{Hom}_A(M, N) \xrightarrow{f^*} \text{Hom}_A(M', N)$$

Proof: Let $h \in \text{Hom}_A(M'', N)$ such that $g^*(h) = 0 \in \text{Hom}_A(M, N)$, i.e., $g^*(h) = h \circ g = 0$, therefore $h \circ g(M) = h(g(M)) = h(M'') = 0$, which means $h = 0 \in \text{Hom}_A(M'', N)$ and thus g^* is injective;

by the exactness, we have $g \circ f = 0$, thus for $\forall h \in \text{Hom}_A(M'', N)$, $(f^* \circ g^*)(h) = f^*(g^*(h)) = (h \circ g) \circ f = h \circ (g \circ f) = 0$, therefore $f^* \circ g^* = 0$ and $\text{Im} g^* \subseteq \ker f^*$. On the other hand, for $\forall h \in \ker f^* \subseteq \text{Hom}_A(M, N)$, we need to prove that $\exists h' \in \text{Hom}_A(M'', N)$

such that $g^*(h') = h' \circ g = h$:

$$\begin{array}{ccccc} & & N & & \\ & & \uparrow h & \nearrow h' & \\ M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' \longrightarrow 0 \end{array}$$

For $\forall x'' \in M''$, $\exists x \in M$ such that $g(x) = x''$, define

$$h' : M'' \rightarrow N$$

$$x'' \mapsto h(x)$$

we show that h' is well-defined: if $x_1, x_2 \in M$ satisfy $g(x_1) = g(x_2) = x''$, then $g(x_1 - x_2) = 0$, i.e., $x_1 - x_2 \in \ker g = \text{Im} f$, hence $\exists x' \in M'$ such that $f(x') = x_1 - x_2$, so $x_1 = x_2 + f(x')$. Thus $h(x_1) = h(x_2 + f(x')) = h(x_2) + (h \circ f)(x')$. Since $h \circ f = 0$, we have $h(x_1) = h(x_2)$, i.e., $h'(x'')$ does not depend on the choice of the preimage of x'' and it is well-defined. Thus we have $\ker f^* \subseteq \text{Im} g^*$, therefore $\ker f^* = \text{Im} g^*$. $\square$

Remark 0.1. :Converse of the propotion is true as well.

Similarly, we have the following proposition

Proposition 0.3. :Let

$$0 \longrightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$$

be an exact sequence of A -modules, M is another A -module. Then we have the exact sequence

$$0 \longrightarrow \text{Hom}_A(M, N') \xrightarrow{f_*} \text{Hom}_A(M, N) \xrightarrow{g_*} \text{Hom}_A(M, N'')$$

Proof: Similar to proposition 0.2 $\square$

Proposition 0.4. :Let

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \xrightarrow{u_1} & M & \xrightarrow{v_1} & M'' \longrightarrow 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' \\ 0 & \longrightarrow & N' & \xrightarrow{u_2} & N & \xrightarrow{v_2} & N'' \longrightarrow 0 \end{array}$$

be a commutative diagram with exact rows. Then the sequence

$$0 \longrightarrow \ker f' \xrightarrow{\bar{u}_1} \ker f \xrightarrow{\bar{v}_1} \ker f'' \xrightarrow{d} \text{coker} f' \xrightarrow{\bar{u}_2} \text{coker} f \xrightarrow{\bar{v}_2} \text{coker} f'' \longrightarrow 0$$

is exact. Here, $\bar{u}_1 = u_1|_{\ker f'}$, $\bar{v}_1 = v_1|_{\ker f}$ and $\bar{u}_2$ is induced by u_2 , $\bar{v}_2$ is induced by v_2 , i.e., $\bar{u}_2(\bar{x}) = \overline{u_2(x)}$ and $\bar{v}_2(\bar{x}) = \overline{v_2(x)}$.

Proof: We define $d : \ker f'' \rightarrow \text{coker} f'$ as follows:

For $\forall x'' \in \ker f'' \subseteq M''$, $\exists x \in M$ such that $v_1(x) \in x''$. Since $f''(v_1(x)) = f''(x'') = 0$, by the comutativity of the diagram, we have $v_2(f(x)) = 0$, i.e., $f(x) \in \ker v_2 = \text{Im} u_2$, therefore $\exists z' \in N'$ such that $u_2(z') = f(x)$. Now define

$$d : \ker f'' \rightarrow \text{coker} f'$$

$$x'' \mapsto \overline{z'} = z' + \text{Im} f'$$

We shall check that d is well-defined:

We only need to prove that $\overline{z'}$ does not depend on the choice of the preimage of x'' .

Suppose $x_1, x_2 \in M$ such that $v_1(x_1) = v_1(x_2) = x''$, hence $v_1(x_1 - x_2) = 0$, i.e., $x_1 - x_2 \in \ker v_1 = \text{Im} u_1$. Thus $\exists y' \in M'$ such that $u_1(y') = x_1 - x_2$, i.e., $x_1 = u_1(y') + x_2$. So $f(x_1) = f(x_2) + f(u_1(y'))$ and $f(x_1 - x_2) = f(u_1(y')) = u_2(f'(y'))$ by the commutativity of the diagram. Since $f(x_1), f(x_2) \in \ker v_2 = \text{Im} u_2$, hence $\exists z_1 \in N'$ such that $u_2(z_1) = f(x_1)$ and $\exists z_2 \in N'$ such that $u_2(z_2) = f(x_2)$, so we have $u_2(z_1 - z_2) = f(x_1 - x_2) = u_2(f'(y'))$. Since u_2 is injective, we get $z_1 - z_2 = f'(y') \in \text{Im} f'$, which means that $z_1 + \text{Im} f' = z_2 + \text{Im} f'$ and d is independent of the choice of the preimage of $x'' \in \ker f''$. Thus d is well-defined.

The exactness of the sequence is not hard to show. $\square$

Additive Function

Definition 0.2. :Let $\mathcal{C}$ be a class of A -modules, a numerical function

$$\lambda : \mathcal{C} \rightarrow \mathbb{N}$$

is said to be **additive** if for any short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

we have $\lambda(M') - \lambda(M) + \lambda(M'') = 0$.

Example 0.3. :Suppose $\mathcal{C}$ is the class of vector spaces. Define

$$\lambda : \mathcal{C} \rightarrow \mathbb{N}$$

$$V \mapsto \dim V$$

then for any short exact sequence of vector spaces

$$0 \longrightarrow V_1 \xrightarrow{\varphi} V_2 \xrightarrow{\psi} V_3 \longrightarrow 0$$

i.e., φ is an injective linear operator and ψ is a surjective linear operator, by linear algebra theory we can know that

$$\dim(\ker \varphi) + \dim(\text{Im} \varphi) = \dim(\text{Im} \varphi) = \dim V_1$$

and

$$\dim(\ker \psi) + \dim(\text{Im} \psi) = \dim(\text{Im} \varphi) + \dim V_3 = \dim V_2$$

Hence we have

$$\dim V_1 - \dim V_2 + \dim V_3 = 0$$

i.e., $\lambda(V_1) - \lambda(V_2) + \lambda(V_3) = 0$ and λ is an additive function.

Proposition 0.5. :Suppose $\mathcal{C}$ is the class of A -modules and $\lambda : \mathcal{C} \rightarrow \mathbb{N}$ is an additive function. Let

$$0 \longrightarrow M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} M_3 \longrightarrow \dots \longrightarrow M_{n-1} \xrightarrow{\varphi_{n-1}} M_n \longrightarrow 0$$

be an exact sequence of A -modules. Then we have

$$\sum_{i=1}^n (-1)^{i-1} \lambda(M_i) = 0$$

Proof: By the long exact sequence, we have the following short exact sequences

$$0 \longrightarrow M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} \text{Im} \varphi_2 \longrightarrow 0$$

$$0 \longrightarrow \text{Im} \varphi_2 \xrightarrow{\varphi_2} M_3 \xrightarrow{\varphi_3} \text{Im} \varphi_3 \longrightarrow 0$$

$$0 \longrightarrow \text{Im} \varphi_3 \xrightarrow{\varphi_3} M_4 \xrightarrow{\varphi_4} \text{Im} \varphi_4 \longrightarrow 0$$

.....

$$0 \longrightarrow \text{Im} \varphi_{n-2} \xrightarrow{\varphi_{n-2}} M_{n-1} \xrightarrow{\varphi_{n-1}} \text{Im} \varphi_{n-1} = M_n \longrightarrow 0$$

hence we have $\lambda(M_1) - \lambda(M_2) + \lambda(\text{Im} \varphi_2) = 0$, $\lambda(\text{Im} \varphi_2) = \lambda(M_3) - \lambda(\text{Im} \varphi_3)$. Therefore, $\lambda(M_1) - \lambda(M_2) + \lambda(M_3) - \lambda(\text{Im} \varphi_3) = 0$, $\lambda(\text{Im} \varphi_3) = \lambda(M_4) - \lambda(\text{Im} \varphi_4)$. By induction, we get $\sum_{i=1}^n (-1)^{i-1} \lambda(M_i) = 0$. $\square$